

PROBUSE CONSULTING SERVICE PVT. LTD.

Odoo Migration Cookbook

Version 19 to Version 20

COMMUNITY EDITION AND ENTERPRISE EDITION · FIRST EDITION · SEPTEMBER 2026

Mr. Mustufa Rangwala

Ms. Sananaz Mansuri

Ahmedabad, Gujarat, India

Website: <https://www.probuse.com>

Website: <https://probuseappdemo.com/>

Email: contact@probuse.com

This cookbook lists the code replacements required when moving custom modules from Odoo 19.0 to Odoo 20.0. It covers **Odoo Community** and **Odoo Enterprise**. Each entry is a 19 → 20 change: a model, field, method, view xpath, QWeb directive, HTTP type, or JavaScript API that 19 accepted and 20 does not.

The replacements are taken from the official Odoo **19.0** and **20.0** source trees (`odoo/odoo` Community, including `base` and the standard addons, and the matching Enterprise addons).

Applies to

Mark	Meaning
CE + EE	Same change in Community 20 and Enterprise 20
CE	Community 20 only, or a Community model that Enterprise does not use
EE	Enterprise 20 only (Payroll, Budget, Field Service, Sign, Enterprise Helpdesk, Enterprise Accounting extras)

Authors

Mr. Mustufa Rangwala

Ms. Sananaz Mansuri

Published by

Probuse Consulting Service Pvt. Ltd.

Ahmedabad, Gujarat, India

Website: <https://www.probuse.com>

Website: <https://probuseappdemo.com/>

Email: contact@probuse.com

Copyright © 2026 Probuse Consulting Service Pvt. Ltd. All rights reserved.

Odoo is a trademark of Odoo S.A. This is an independent publication. It is not published, endorsed, or affiliated with Odoo S.A.

Confirm each replacement against the Odoo 20.0 Community or Enterprise source you deploy.

About the publisher

Probuse Consulting Service Pvt. Ltd. is an Odoo consulting and product company based in Ahmedabad, Gujarat, India, founded on 1 November 2014. The firm implements, customizes, and supports Odoo for customers worldwide.

Website: <https://www.probuse.com>

Website: <https://probuseappdemo.com/>

Email: contact@probuse.com

About the authors

Mr. Mustufa Rangwala is Founder, CEO, and Managing Director of Probuse Consulting Service Pvt. Ltd. He has worked in the Odoo / OpenERP ecosystem since the early versions and previously worked with Odoo India.

Ms. Sananaz Mansuri is a Consultant and Odoo techno-functional implementer. She works on requirement analysis, process mapping, configuration, and delivery. She is the GCC contact for Probuse Consulting Service Pvt. Ltd.

Contents

Select a chapter to open it.

1. Security

2. ORM and Python

3. Search and domains

4. Binary, Excel, PDF, and time

5. Configuration, HTTP, and mail

6. SQL reports

7. Views

8. QWeb and reports

9. Owl 3

10. Accounting and payments

11. Sales, purchase, and products

12. Inventory, MRP, and repair

13. HR, time off, and payroll

14. Project, timesheets, and portal

15. CRM, calendar, and mail

16. Website and eCommerce

17. Enterprise budget, Field Service, Helpdesk, and Fleet

18. Symptom index

19. Mapping tables

1. Security

[BACK TO CONTENTS](#)

CE + EE

Odoo 20 loads `ir.access.csv`. It does not load `ir.model.access.csv` plus XML `ir.rule` records. XML `<record model="ir.rule">` raises `KeyError: 'ir.rule'`.

Odoo 19

```
id,name,model_id:id,group_id:id,perm_read,perm_write,perm_create,perm_unlink
access_lock,lock,model_my_lock,base.group_user,1,1,1,0
```

Odoo 20 — `security/ir.access.csv`

```
id,name,model_id,group_id/id,operation,domain
access_my_lock,my.lock,my.lock,base.group_user,cru,
```

Column	Odoo 20
<code>model_id</code>	Technical model name (<code>sale.order</code>). Not <code>module.model_x</code> .
<code>group_id/id</code>	Group XML id. Empty = global rule.
<code>operation</code>	<code>crud</code> , <code>cru</code> , <code>cr</code> , ... — not <code>rc</code> , and not <code>perm_read</code> columns
<code>domain</code>	Record-rule domain, or empty

Further 19 → 20 rules on this file:

- `model_id/id` from 19 is not `model_id` . The 20 column needs the technical name.
- Domains `[(1, '=', 1)]` and `[('1', '=', 1)]` fail validation. Use an empty domain.

- A domain on a non-stored field raises `Field has no SQL representation`. Use a stored field or a stored path.
- Company domain `('company_id', 'in', company_ids + [False])` fails. Use:

```
[|'|, ('company_id', '=', False), ('company_id', 'in', company_ids)]
```

- Custom `res.groups` XML must be listed in the manifest **before** `ir.access.csv` if the CSV references those groups.
-

2. ORM and Python

[BACK TO CONTENTS](#)

CE + EE unless marked otherwise.

Odoo 19	Odoo 20
<code>_sql_constraints = [('name', 'sql', 'msg')]</code>	<code>_name = models.Constraint(</code>
<code>self.fields_view_get(...)</code>	<code>self._get_view(...)</code>
<code>self.check_access_rights('read')</code>	<code>self.check_access('read')</code>
<code>self.check_access_rights('read', raise_exception=False)</code>	<code>self.has_access('read') (b</code>
<code>self.user_has_groups('sales_team.group_sale_manager')</code>	<code>self.env.user.has_groups('</code>
<code>self._context</code>	<code>self.env.context</code>
<code>self.env.user.company_id</code>	<code>self.env.company</code>
<code>ir.config_parameter.get_param / set_param</code>	<code>get_int / get_str / get_bc set_str / set_bool / set_f</code>
<code>pytz</code>	<code>datetime.UTC and zoneinfo</code>
<code>window action toggle_active</code>	<code>action_archive / action_ur</code>
<code>write archive=True</code>	<code>write active=False</code>
<code>mail.thread.cc</code>	<code>mail.thread</code>

```
# 19
_sql_constraints = [
    ('code_uniq', 'unique(code, company_id)', 'Code must be unique per company.'),
]

# 20
_code_uniq = models.Constraint(
    'unique(code, company_id)',
    'Code must be unique per company.',
)
```

```
# 19
self.check_access_rights('read')

# 20
self.check_access('read')
self.has_access('read')
```

```
# 20 – selection_add requires ondelete
state = fields.Selection(selection_add=[
    ('extra', 'Extra'),
], ondelete={'extra': 'set default'})
```

`_name` listed in `_inherit` requires that parent model to exist in the registry. CE a module that `_name = 'helpdesk.sla'` and also `_inherit = 'helpdesk.sla'` fails unless EE `helpdesk` is installed. CE-only graphs use `_name` only. EE graphs that load Enterprise Helpdesk keep `_inherit`.

`mail.thread.cc` is removed. A model that inherits `rating.mixin` already inherits `mail.thread`. Listing both on 20 raises an MRO error.

```
# 20
_inherit = ['portal.mixin', 'mail.activity.mixin', 'rating.mixin']
```

Two Many2many fields on the same model to the same comodel, without `relation=`, collide on 20. Set `relation=` on at least one.

A 20 compute must assign every record. A 20 compute must not return a value dict. Stored and non-stored fields cannot share one compute method.

`method=True` on a computed field is invalid. `tracking=` on a related field requires `mail.thread`. Related Selection must not re-declare `selection / default / tracking.placeholder=` is XML-only.

A required SQL column without `default=` raises `NOT NULL` on create.

`required=True` on a new `Many2one` cannot add `NOT NULL` when the table already has rows.

```
self.env['some.model'] is a recordset. _inherit is on
self.env.registry['some.model'] .
```

COPY, WRITE ORDER, AND TRANSLATIONS

A Char field named `name` now defaults to `copy=mark_as_copy('name')`.

Duplicating a record appends “(copy)” to that value. Override `copy=` when the 19 behaviour must stay a plain copy.

```
from odoo.tools.translate import mark_as_copy

name = fields.Char(copy=mark_as_copy('name'))
code = fields.Char(copy=False)
```

`create()` and `write()` run field inverses in (`field.write_sequence`, `field.definition order`). Custom inverses that assumed 19 order can break. Set `write_sequence=` when two inverses must run in a fixed sequence.

```
partner_id = fields.Many2one('res.partner', write_sequence=10)
```

Reading a `One2many` or `Many2many` returns **only records the user can access**. Code that counted or mapped the full relational cache of forbidden ids will see a shorter recordset.

Module translations are stored in **that module's PO**. 20 no longer dumps them into `base`. Export and import translations per module.

WEBSITE ON THE ENVIRONMENT

CE + EE when `website` is installed. 19 model code often used `request.website`.
20 Environment exposes `env.website` from `context['website_id']`. Prefer
`self.env.website` in models. `request` is not bound in report parsers and
several ORM contexts (Chapter 4).

```
# 20
website = self.env.website
```

3. Search and domains

[BACK TO CONTENTS](#)

CE + EE

Odoo 19	Odoo 20
<code>_name_search</code>	<code>_rec_names_search</code> and/or <code>_search_display_name</code>
<code>from odoo.osv import expression / expression.AND / expression.OR</code>	<code>from odoo.fields import Domain / Domain.AND / Domain.OR</code>
<code>from odoo.fields import Domain, AND, FALSE_DOMAIN</code>	<code>from odoo.fields import Domain only (Domain.FALSE)</code>
<code>_name_search(..., args=..., name_get_uid=...)</code>	<code>(self, name, domain=None, operator='ilike', limit=None, order=None)</code>
<code>_read_group_stage_ids(self, stages, domain, order)</code>	<code>_read_group_stage_ids(self, stages, domain)</code>
<code>view modifier in 'x' / not in 'x'</code>	<code>== 'x' / != 'x' (tuple in ('a', 'b') is valid)</code>

```
# 20
class AccountAnalyticAccount(models.Model):
    _inherit = 'account.analytic.account'
    _rec_names_search = ('name', 'code', 'custom_number', 'partner_id')
```

```
from odoo.fields import Domain

def _search_display_name(self, *args, **kwargs):
    return Domain(super()._search_display_name(*args, **kwargs))
```

`odoo.tools.Query` is not exported from `odoo.tools`. Import `SQL` from `odoo.tools`.

DOMAIN OPERATOR ACCESS

`20` adds the domain operator `access`. It is valid on `id` and on `Many2one` fields. The value is `'read'`, `'write'`, `'create'`, or `'unlink'`. Use it when a domain must keep only records (or related records) the user may operate on. It does not work on properties.

```
# 20 - current user can write the partner
Domain('partner_id', 'access', 'write')

# 20 - current user can read this record
Domain('id', 'access', 'read')
```

This operator is useful in `ir.access.csv` domains (Chapter 1) when the rule must follow comodel rights.

4. Binary, Excel, PDF, and time

BACK TO CONTENTS

CE + EE

BINARY

On the record, a Binary field is `BinaryValue (odoo.tools.binary)`. `.content` is already raw bytes. `base64.decodebytes(self.datas)` double-decodes or raises `Incorrect padding`.

Direction	Odoo 20
Read	<code>value.content</code> or <code>bytes(value)</code>
Filename	<code>value.filename</code> — optional name on <code>BinaryValue</code> / <code>BinaryBytes</code>
Write	<code>from odoo.tools.binary import BinaryBytes</code> — not raw <code>bytes</code> , not <code>base64.b64encode(...)</code>
Image write	<code>image_apply_opt(pil_image, 'PNG')</code> then <code>BinaryBytes(...)</code> . <code>image_to_base64</code> is removed
<code>ir.attachment</code>	field <code>raw</code> , not <code>datas</code> ; <code>res_id</code> is an integer
XML binary	<code>type="bytes"</code>
RPC / HTTP	binary payload is <code>BinaryValue</code> , not a bare base64 string

`BinaryBytes(data, filename=name)` keeps the original file name on the value. RPC and HTTP that still send a raw base64 string without that wrapper fail the 20 type check.

```

from odoo.tools.binary import BinaryBytes
from odoo.tools.image import image_apply_opt

record.image_1920 = BinaryBytes(image_apply_opt(pil_image, 'PNG'))
record.datas = BinaryBytes(stream.getvalue(), filename=name)

```

```

<!-- 19 -->


<!-- 20 -->
<br><code>mail_post_autofollow</code> |
| EE server action <code>use_in_ai</code>                         | <code>ai_tool_name</code> matching <code>[a-zA-Z0-9_]{1,64}</code>                           |
| settings inherit<br><code>group[@name='preferences']</code>     | <code>//page[@name='preferences']//group[@name='other_prefe</code>                           |
| employee settings xpath                                         | after <code>employees_setting_container</code>                                               |

A mail template that inherits an Enterprise model must use the **defining** module’s model xmlid. Example EE: `account_budget.model_budget_analytic` . Inherit does not create `your_module.model_budget_analytic` .

## MAIL.TRACKING.VALUE

CE + EE. 19 stores `mail.tracking.value` in core `mail` . 20 moves the model to the Community addon `mail_tracking` (Discuss Tracking). Core `mail` still generates tracking messages; it does not keep the stored tracking rows unless `mail_tracking` is installed.

| Odoo 19                                                             | Odoo 20                                                                              |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <code>mail.tracking.value</code> in <code>mail</code>               | install <code>mail_tracking</code> ; then <code>_name = 'mail.tracking.value'</code> |
| search <code>mail.tracking.value</code> with only <code>mail</code> | empty / missing model unless <code>mail_tracking</code> is in the graph              |

Depend `mail_tracking` when custom code reads stored old/new values.

### EXTERNAL RPC

`/json/2/<model>/<method>` (JSON-2, `type='json2'` , bearer auth) is the documented external API on 20. The same controller already exists on 19. XML-RPC ( `/xmlrpc/2` ) is still present on both versions. New integrations should call JSON-2. Do not treat XML-RPC as removed in 20.

```
POST /json/2/res.partner/search_read
```

Controller routes that the web client calls stay `type='jsonrpc'` (table above). That is not the same as JSON-2.

---

## 6. SQL reports

---

[BACK TO CONTENTS](#)

### CE + EE

#### `_TABLE_QUERY` → `_TABLE_SQL`

19 SQL views set `_auto = False` and define `_table_query` (attribute or method).

20 **removes** `_table_query`. The lint on 20 is: overwrite `_table_sql`. Return `odoo.tools.SQL`.

```
19
_auto = False

def _table_query(self):
 return self._query()

20
from odoo.tools import SQL

_auto = False

def _table_sql(self) -> SQL:
 return SQL("(%s)", self._query())
```

`sale.report` on 20 is the reference: `_table_sql()` builds a `query` and returns `query.subselect(...)`. A module that only migrated `_select()` / `_group_by()` to `SQL` still fails if it leaves `_table_query` in place.

#### `_SELECT()` AND `_GROUP_BY()`

`_select()` and `_group_by()` return `odoo.tools.SQL`. A Python string cannot be concatenated onto that value. `LiteralsQL` has no `.code`.

```

from odoo.tools import SQL

def _select(self):
 return SQL("%s,\n t.custom_type", super()._select())

```

| Report                      | Odoo 20 hook                                                         |
|-----------------------------|----------------------------------------------------------------------|
| report.project.task.user    | <code>_select()</code> / <code>_group_by()</code> → <code>SQL</code> |
| timesheets.analysis.report  | <code>_select()</code> → <code>SQL</code>                            |
| sale.report                 | <code>_select_dict</code> + <code>_groupby_list</code>               |
| purchase.report             | <code>_select_list(table)</code> + <code>_groupby_list(table)</code> |
| account.move invoice report | <code>_select_list</code>                                            |
| report.pos.order            | wrap <code>_select</code> only                                       |

A field added on `analytic.view_account_analytic_line_filter` must also be added on `timesheets.analysis.report` through `_select()`.

Translated `name` in SQL:

```
COALESCE(project_task_type.name->>'en_US', project_task_type.name::text)
```

# 7. Views

BACK TO CONTENTS

CE + EE unless marked otherwise.

## KANBAN

| Odoo 19 inherit                                  | Odoo 20 inherit                                                                               |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <code>project.view_project_kanban</code><br>body | <code>project.view_project_card</code>                                                        |
| <code>project.view_task_kanban</code> footer     | <code>project.view_task_card</code> <code>//t[@t-</code><br><code>name='card']//footer</code> |
| kanban template name                             | <code>t-name="card"</code>                                                                    |

`project.view_task_kanban` `USES` `card_id="% (project.view_task_card)d"` .

## RIBBONS AND BUTTON ICONS

```
<!-- 19 -->
<button icon="fa-check"/>
<widget name="web_ribbon" text="Archived" bg_color="bg-info"/>

<!-- 20 -->
<button icon="check"/>
<widget name="web_ribbon" title="Archived" bg_color="text-bg-info" invisible="acti
```

Odoo 19	Odoo 20
<code>web_ribbon</code> <code>text=</code>	<code>title=</code>
<code>bg_color="bg-info" / bg-primary</code>	<code>bg_color="text-bg-info" / text-bg-primary / text-bg-success / text-bg-warning / text-bg-danger</code>
<code>button</code> <code>icon="fa-check"</code>	<code>icon="check"</code> — names from <code>web/icons.py</code> <code>ICONS</code>
Font Awesome / GTK icon names on <code>ViewButton</code>	do not draw. <code>iconFromString</code> renders <code>oi</code> + <code>data-icon=</code>
Font Awesome on website snippets	Material Symbols ligature (same names as <code>ViewButton</code> )

Legal `ViewButton` names include `check`, `save`, `close`, `history`, `edit_square`, `task`, `group`, `book`, `description`, `stethoscope`.

Owl 3 `DropDownItem` takes `attrs=`, `href=` and `dataset=` are invalid props.

## XPATH

`View inherit` cannot use attribute `string` or `title` as a selector. Use `hasClass(...)`, `name`, `id`, or `t-call`.

Xpath cannot enter the body of a `t-call`. Inherit the called template, or xpath the `t-call` node.

`[@class='X']` is an exact match. Convert the inherit only when the 20 parent class string no longer equals `X`. Use `hasClass('X')` in that case.

Invoice, sale, and purchase order line `product_id / name` are inside `<column name="product_and_description"> . /list/field[@name='product_id']` does not match.

Xpath a direct list child ( `quantity`, `product_qty`, `analytic_distribution`, `account_id` ).

Form	Odoo 19 xpath	Odoo 20
Analytic account	<code>//header</code>	no header — insert <code>&lt;header&gt;</code> w
Timesheet	<code>add button_box</code>	<code>div[@name='button_box']</code> alre
Calendar event	<code>button_box</code>	no <code>button_box</code> — buttons in <code>/</code> <code>//sheet/widget[@name='web_r</code>
Employee	<code>oe_title</code>	<code>//h1</code> or <code>//h1[hasclass('o_er</code>
Job	<code>user_id</code>	after <code>department_id</code> ; form alre
CRM	<code>//div[hasclass('oe_title')]/div</code>	<code>//div[hasclass('oe_title')]</code>
Maintenance request	<code>archive_equipment_request</code> / <code>user_id</code>	<code>//header</code> ; assignees <code>user_ids</code>
Invoice totals	<code>@class='oe_subtotal_footer'</code>	<code>//group[hasclass('oe_subtot</code>
Equipment	19 <code>location_id</code>	<code>owner_user_id</code>
Partner list	<code>complete_name</code>	<code>display_name</code>
Partner search	<code>group[@name='group_by']</code>	<code>//search/group</code>
User preferences	<code>group[@name='preferences']</code>	<code>//page[@name='preferences']</code>
Product	<code>product.product_variant_easy_edit_view</code>	removed
Product action	<code>product.product_normal_action</code>	<code>product.product_variant_act</code>
Sale config menu	<code>sale.prod_config_main</code>	<code>sale.menu_sale_config</code>
Project form	<code>group[@name='extra_settings']</code>	<code>//page[@name='settings']</code>
Timesheet kanban footer	<code>footer[@class='mt-0 pt-0']</code>	<code>div[@name='ts_footer']</code>
eLearning channel	<code>//field[@name='active']</code>	no <code>active</code> — xpath <code>name</code>

Form	Odoo 19 xpath	Odoo 20
Activity popup	<code>company_id</code>	after <code>summary</code>
POS session	<code>move_id</code>	<code>user_id</code>
POS receipt	Owl <code>ReceiptHeader</code> / <code>pos-receipt-contact</code>	<code>point_of_sale.pos_order_rec</code>
Sale form	<code>show_update_fpos</code>	<code>fiscal_position_id</code>
Sale portal sidebar	<code>navspy</code>	<code>div#sale_order_sidebar_butt</code>

`product.product_variant_action` `context` `active_id` is defined only on a product form. A custom `ir.actions.act_window` copied from that action must not keep `active_id` if it opens from another model.

---

## 8. QWeb and reports

[BACK TO CONTENTS](#)

### CE + EE

`<act_window>` and `<report>` are not in Odoo 20 `import_xml.rng`. Use `ir.actions.act_window` and `ir.actions.report` records. `ir.actions.report` has no `report_file`.

```
<record id="action_print_doc" model="ir.actions.report">
 <field name="name">Document</field>
 <field name="model">sale.order</field>
 <field name="report_type">qweb-pdf</field>
 <field name="report_name">my_module.report_document</field>
 <field name="binding_model_id" ref="sale.model_sale_order"/>
 <field name="binding_type">report</field>
</record>
```

Odoo 19	Odoo 20
<code>t-esc</code>	<code>t-out</code>
<code>t-raw</code>	<code>t-out</code>
<code>t-field-options</code> / <code>t-esc-options</code>	<code>t-options</code>
<code>t-field</code>	still valid
invoice <code>o.type</code>	<code>o.move_type</code>
BoM overview header <code>BoM Cost</code>	<code>Total Cost</code>
<code>from odoo.tools import</code> <code>html2plaintext</code>	<code>from odoo.tools.mail import</code> <code>html2plaintext</code>
<code>web_editor</code> Python <code>get_video_embed_code</code>	package removed; embed is JS <code>html_editor</code>

An xpath that selected @t-esc must select @t-out .

REPORT ENGINES

CE + EE. 20 splits PDF rendering into two Community modules that 19 did not ship as addons:

Engine	Odoo 20 module
wkhtmltopdf	base_report_wkhtmltox
Paper Muncher (in-house)	base_report_paper_muncher

A custom report that assumed “one wkhtml pipeline inside web ” must depend on the engine actually installed. Paper Muncher is the 20 in-house renderer ( <https://odoo.github.io/paper-muncher/> ).

EE If the report bound to hr.work.entry rows: those document records are gone. Bind to hr.payslip and iterate worked\_days\_line\_ids .

---

## 9. Owl 3

[BACK TO CONTENTS](#)

### CE + EE

Odoo 20 web is Owl 3.

Odoo 19	Odoo 20
<code>import { useState } from "@odoo/owl"</code>	<code>import { proxy } from "@odoo/owl" then this.state = proxy({ ... })</code>
<code>template state.foo / kpis.foo / dashboardData.foo</code>	<code>this.state.foo / this.kpis.foo / this.dashboardData.foo / this.props.foo</code>
<code>t-esc</code>	<code>t-out</code>
<code>t-model</code> on a <code>proxy()</code> field	<code>t-model</code> accepts a signal only. <code>proxy()</code> fields use <code>t-att-value</code> + <code>t-on-input</code>
<code>t-set</code> inside <code>t-call</code> as a sub-template variable	slot body — pass parameters as attributes on <code>&lt;t t-call="..." /&gt;</code>
<code>standardActionServiceProps</code> from <code>action_service</code>	<code>@web/webclient/actions/action_plugin</code>
Chart.js v2 options	Chart.js v4 ( <code>cutout</code> , <code>plugins.legend</code> , <code>plugins.title</code> , <code>scales.y.beginAtZero</code> ) at <code>web/static/lib/Chart/Chart.js</code>
<code>_attachment_format</code>	<code>_store_attachment_fields</code>
<code>mail.AttachmentList</code> props <code>attachments</code>	<code>attachmentGroups</code> ; container <code>o-mail-AttachmentContainer</code>
<code>mail.assets_messaging</code>	empty — put JS/XML in <code>web.assets_backend</code>
<code>window.alert</code> in systray	notifications; company is <code>user.activeCompany</code>

```
import { proxy } from "@odoo/owl";
this.state = proxy({ total: 0 });
```

```
<t t-out="this.state.total"/>
<t t-out="this.kpis.open"/>
```

`t-as` loop locals stay unprefixed. A static CSS class is not a template identifier ( `ud-widget` stays `ud-widget` ).

```
static props = { "*": true };
```

```
import { signal } from "@odoo/owl";
this.searchTerm = signal("");
```

Mail store **CE + EE**:

```
Store().add(channel, "_store_channel_fields")._build_result()
```

```
this.store["mail.thread"].getOrFetch({ model: "discuss.channel", id })
```

A Python method called from Owl as `[[], payload]` must be `@api.model`.

Owl 3 also drops or relocates several 19 hooks. Replace them when the 20 component fails to compile or never re-renders.

Odoo 19	Odoo 20
<code>this.render()</code>	the <code>proxy</code> / <code>signal</code> update re-renders; do not call <code>render</code>
<code>t-portal</code>	removed — render into the target node, or a 20 portal helper if the standard component still exposes one
<code>onWillUpdateProps</code> / <code>onWillRender</code> / <code>onRendered</code>	removed from Owl 3. Use <code>onWillStart</code> , <code>onMounted</code> , <code>onWillUnmount</code> , and <code>onPatched</code>

# 10. Accounting and payments

---

[BACK TO CONTENTS](#)

**CE + EE unless marked EE.**

Odoo 19	Odoo 20
invoice lines <code>journal_line_ids</code>	<code>line_ids</code>
invoice salesperson group	<code>sale_info_group</code> (customer invoices)
vendor bill extra fields	<code>accounting_info_group</code>
journal entry other info	<code>other_tab_entry</code>
header dates / flags	<code>header_right_group</code>
<code>res.partner.bank.acc_number</code>	<code>account_number</code>
<code>res.partner.bank.acc_holder_name</code>	<code>holder_name</code>
<code>res.partner.bank.bank_id.name</code> / <code>bank_id.bic</code>	<code>bank_name</code> / <code>bank_bic</code>
<code>account.journal.bank_id</code>	<code>bank_name</code>
<code>res.bank</code>	removed — Char, or <code>res.partner.bank</code>
<code>account.analytic.account.group_id</code>	<code>plan_id</code>
<code>payment.provider.state</code>	<code>is_live</code> / <code>is_published</code>
<code>payment.controllers.post_processing</code>	removed
payment transaction group <code>//group//group</code>	<code>//group[@name='transaction_details']</code>
invoice report <code>o.type</code>	<code>o.move_type</code>
<code>account.group</code>	removed
<code>account.account.group_id</code>	<code>parent_id</code> / <code>parent_ids</code> on <code>account.account</code>
account <code>code</code> required	<code>code</code> is computed / optional ( <code>code_store</code> , <code>placeholder_code</code> )
<code>account.payment</code> state <code>in_process</code>	<code>paid</code>
<code>account.payment</code> state <code>paid</code>	<code>reconciled</code>

## ACCOUNT GROUPS → PARENT ACCOUNTS

19 has `account.group` and `account.account.group_id`. 20 deletes `account.group`.  
Hierarchy lives on `account.account`:

```
20
parent_id = fields.Many2one('account.account', string="Parent Account")
parent_ids = fields.Many2many('account.account', compute='_compute_parent_ids')
```

Replace searches on `account.group` and writes to `group_id` with parent accounts. Account codes are no longer a required stored Char; 20 computes code ( `compute_sql` ). Reports that JOIN a mandatory `account_account.code` must tolerate empty codes and use `placeholder_code` when displaying.

## PAYMENT STATES

Odoo 19	Odoo 20
draft	draft
in_process	paid
paid	reconciled
canceled	canceled

Filters, automations, and domains that still use `in_process` match nothing. A payment that 19 called “Paid” is **Reconciled** on 20. A payment that 19 called “In Process” is **Paid** on 20.

## FIXED ASSETS — EE

19 Enterprise treats a depreciation template as `account.asset` with `state == 'model'`. 20 Enterprise adds `account.depreciation.model`. Bind templates, default depreciation methods, and CoA-generated models to that model. `account.asset` remains the asset document.

`payment.method.provider_id` is required. Create the provider first. There is no `payment.payment_method_card xmlid` in `payment`.

Custom fields copied from payment to the journal entry go in

`_generate_move_vals` / `_synchronize_to_moves`.

Invoice line `_compute_account_id` assigns `line.account_id`. It does not return.

---

# 11. Sales, purchase, and products

[BACK TO CONTENTS](#)

## CE + EE

Odoo 19	Odoo 20
<code>sale.order</code> state <code>done</code>	removed. Valid: <code>draft</code> / <code>sent</code> / <code>sale</code> / <code>cancel</code>
<code>purchase.order.line</code> UoM <code>product_uom_id</code>	<code>uom_id</code>
<code>sale.order.line</code> / <code>account.move.line</code> UoM	still <code>product_uom_id</code>
<code>product.type == 'product'</code>	<code>is_storable</code>
<code>product.pricelist.item._compute_price(..., date)</code> positional	<code>date=</code> keyword ( <code>_compute_base_price</code> the same)
<code>sale.prod_config_main</code>	<code>sale.menu_sale_config</code>
<code>product.product_normal_action</code>	<code>product.product_variant_action</code>
<code>product.product_variant_easy_edit_view</code>	removed

`sale.order.line` still has `product_uom_qty` and `product_uom_id`.

## 12. Inventory, MRP, and repair

[BACK TO CONTENTS](#)

### CE + EE

Odoo 19	Odoo 20
<code>stock.scrap</code>	<code>stock.move</code> with <code>is_scrap=True</code>
<code>scrap</code> <code>name</code>	<code>reference</code>
<code>scrap / move</code> UoM <code>product_uom</code> / <code>product_uom_id</code>	<code>uom_id</code>
<code>scrap</code> <code>dest</code> <code>scrap_location</code>	<code>company.scrap_location_id</code>
<code>scrap</code> <code>validate</code>	<code>action_scrap()</code>
<code>scrap</code> <code>form</code> / <code>action</code>	<code>stock.view_scrap_move_form</code> / <code>stock.stock_scrapped_moves_action</code>
<code>stock.return.picking</code>	<code>stock.picking._create_return()</code>
<code>stock.quant.package</code>	<code>stock.package</code>
<code>stock_picking_batch</code>	merged into <code>stock</code> ; inherit <code>stock.stock_picking_batch_form</code>
<code>mrp.bom</code> / <code>mrp.bom.line</code> / <code>repair.order</code> UoM	<code>uom_id</code>
<code>picking</code> <code>move</code> <code>list</code> UoM	<code>//field[@name='move_ids']/list/field[@name='uom_</code>
<code>mrp.production.show_produce_all</code>	removed
MO button “Produce All” (separate from Produce)	one <code>button_mark_done</code>

19 manufacturing orders expose `show_produce_all` and two header buttons (Produce vs Produce All). 20 removes that field. Close the order with

button\_mark\_done only. Xpaths and automations that test show\_produce\_all fail.

---

# 13. HR, time off, and payroll

---

[BACK TO CONTENTS](#)

**CE + EE for employee / time off. EE for Payroll.**

Odoo 19	Odoo 20
<code>hr.contract</code>	<code>hr.version</code>
SQL join <code>hr_contract</code>	<code>hr_version</code>
<code>hr.version.wizard</code>	<code>employee.contract_template_id +</code> <code>_onchange_contract_template_id()</code>
<code>hr.leave.type</code>	<code>hr.work.entry.type</code>
<code>hr.leave.holiday_status_id</code>	<code>work_entry_type_id</code>
<code>hr.leave.action_confirm</code>	removed; <code>action_approve</code> remains
<code>hr.leave.holiday_type</code>	<code>validation_type</code>
inherit time-off type form	<code>hr_holidays.hr_work_entry_type_view_for</code> <code>support_document</code>
<code>hr.expense.sheet</code>	<code>hr.expense</code>
<code>hr_work_entry.work_entry_type_attendance</code>	<code>hr_work_entry.generic_work_entry_type_a</code>
legal leave default	<code>hr_work_entry.generic_work_entry_type_1</code>
leave XML on <code>hr.leave.type</code>	merge onto <code>hr.work.entry.type</code> ( <code>count_as='absence'</code> , <code>time_off_selectabl</code>
depend <code>hr_work_entry_holidays</code> only	<code>hr_holidays</code>
EE <code>hr.payslip.input.type</code>	removed; <code>input_usage_payslip</code> on the salar
EE salary rule <code>category_id</code> / <code>struct_id</code>	<code>category_ids</code> / <code>struct_ids</code> [(6, 0, [ref
EE <code>payslip.paid_amount</code>	<code>payslip._get_basic_salary()</code>

Odoo 19	Odoo 20
EE payment wizard <code>export_format</code> <code>selection_add</code>	<code>_get_export_format_selection()</code>
EE pay-run kanban	<code>hr_payroll.hr_payrun_payslip_kanban_cal</code>
EE <code>hr.work.entry</code> list documents	removed; print from <code>hr.payslip</code> <code>worked_days_line_ids</code>

Search filter `approve` on time off is nested. `//search/filter[@name='approve']` does not match. Use `//filter[@name='approve']` .

Several 20 workflows use state `canceled` , not `cancel` .

---

## 14. Project, timesheets, and portal

[BACK TO CONTENTS](#)

CE + EE unless marked EE.

Odoo 19	Odoo 20
<code>project.task.user_id</code>	<code>user_ids</code>
<code>access domain task_id.user_id</code>	<code>('task_id.user_ids', 'in', [user.id])</code>
<code>note.note</code>	<code>project.task (project_todo)</code>
<code>todo default_memo</code>	<code>default_name (project.task has no memo)</code>
<code>project.portal_my_home</code>	<code>portal.entry (project.portal_projects, project.portal_tasks)</code>
<code>portal card active</code>	<code>show_in_portal</code>
<code>portal_client_category / portal_service_category</code>	<code>removed</code>
<code>portal.portal_sidebar //t[@t-set='entries']</code>	<code>//div[class='col-lg-3']</code>
<code>invoice portal download</code>	<code>//div[class='o_download_pdf']</code>
<code>portal docs wrap</code>	<code>div.o_portal_docs</code>
<code>get_records_pager</code>	<code>removed</code>

`_prepare_home_portal_values` counters are merged in `counters()`. Group-restricted cards use `is_config_card` and `_filter_visible_portal_cards`.

### PROJECT PROFITABILITY — CE

19 Community `project.project` implements `_get_profitability_items`, `_get_profitability_values`, `_get_profitability_labels`, and related helpers. Those

methods are **not** on `Community 20 project` . Overrides and tests that call them raise `AttributeError` . Enterprise 20 reports margin and budget through the budget apps (Chapter 17), not through those 19 CE hooks.

**EE Field Service:** `industry_fsm` is not in Enterprise 20. Depend `project` (and `sign` if signatures are used).

Odoo 19 EE	Odoo 20
<code>industry_fsm</code>	<code>project</code>
<code>industry_fsm.fsm_menu_root</code>	<code>project.menu_main_pm</code>
<code>industry_fsm.group_fsm_user</code>	<code>project.group_project_user</code>

`is_fsm` is not a standard 20 field. Custom domains that still use it need a local Boolean on `project.project` / `project.task` . Enterprise Planning ( `planning.slot` ) is a different model.

---

# 15. CRM, calendar, and mail

[BACK TO CONTENTS](#)

## CE + EE

Odoo 19	Odoo 20
CRM <code>//div[hasclass('oe_title')]/div</code>	<code>//div[hasclass('oe_title')]</code>
lead <code>currency_id</code> (standard)	<code>company_currency</code>
notebook <code>page[@name='lead']</code>	<code>internal_notes</code> / <code>extra</code>
Extra Info <code>company_id</code> in <code>categorization</code>	<code>//page[@name='extra']//field[@name='company_id']</code> ( <code>group[@name='Misc']</code> )
graph amount	<code>type="measure"</code>
calendar <code>button_box</code>	removed — see <a href="#">Chapter 7</a>
calendar <code>page_invitations</code>	<code>notes</code> / <code>options</code>
<code>mail.tracking.value</code> in <code>mail</code>	Community addon <code>mail_tracking</code> ( <a href="#">Chapter 5</a> )

# 16. Website and eCommerce

---

[BACK TO CONTENTS](#)

## CE + EE

```
<!-- 19 - 500 on 20 public pages -->
<input t-attf-value="#{user_id.partner_id.email}"/>

<!-- 20 -->
<input t-att-value="request.env.user.partner_id.email if not request.env.user._is_
```

Odoo 19	Odoo 20
<code>a[@id='add_to_cart']</code>	<code>button[@name='add_to_cart']</code>
add-to-cart wrap	<code>div[@id='add_to_cart_wrap']</code> / <code>website_#o_wsale_cta_wrapper</code>
<code>product._is_add_to_cart_possible()</code> form	removed
<code>website_sale.product_terms_and_conditions</code>	removed — xpath after <code>#o_wsale_product_c</code>
<code>#wrap.js_sale</code>	removed
<code>website_sale_wishlist</code> / <code>website_sale_comparison</code>	merged into <code>website_sale</code>
wishlist / compare xmlids	<code>website_sale.product_wishlist</code> , <code>website_sale.model_product_wishlist</code>
comparison controller	<code>odoo.addons.website_sale.controllers.c</code>
hide price	<code>combination_info['hide_price']</code>
category form <code>sequence</code>	list only — xpath <code>parent_id</code>
<code>res.partner.company_type</code>	removed — <code>is_company</code>
<code>res.partner.mobile</code>	removed — <code>phone</code>
<code>res.users.groups_id</code>	<code>group_ids</code>
action context <code>default_company_type</code>	<code>default_is_company</code>
<code>ir.ui.view.groups_id</code>	unchanged (view access, not the user field)
Font Awesome class on website snippets ( <code>fa fa-...</code> )	Material Symbols ligature / <code>oi</code> + <code>data-icon</code>
<code>request.website</code> in a model method	<code>self.env.website</code> (Chapter 2)

Website and snippet markup that still ships Font Awesome class names will not draw on 20. Use the same ViewButton / Material Symbols names as Chapter 7 ( `check` , `close` , ...). Public pages that read `request` from a model or report parser hit `request` not bound ; use `env.website` .

```
19
vals['company_type'] = 'person'
vals['mobile'] = '...'
user_vals['groups_id'] = [(6, 0, ids)]

20
vals['is_company'] = False
vals['phone'] = '...'
user_vals['group_ids'] = [(6, 0, ids)]
```

---

# 17. Enterprise budget, Field Service, Helpdesk, and Fleet

[BACK TO CONTENTS](#)

## BUDGET — EE

Odoo 19 Enterprise	Odoo 20 Enterprise
<code>crossovered.budget</code>	<code>budget.analytic</code>
<code>crossovered.budget.lines</code>	<code>budget.line</code>
lines field	<code>budget_line_ids</code>
amount field	<code>budget_amount</code>
states	<code>draft</code> / <code>confirmed</code> / <code>revised</code> / <code>done</code> / <code>canceled</code>
form	<code>account_budget.view_budget_analytic_form</code>
action	<code>account_budget.act_budget_analytic_view</code>
report binding	<code>account_budget.model_budget_analytic</code>
<code>committed_amount</code>	on <code>account_budget_purchase</code>

CE Community 20 has no `budget.analytic`. A Community budget module that still uses its own model is not the Enterprise model and is not retargeted to `account_budget`.

## HELPDESK

EE Enterprise `helpdesk` remains. CE a custom `helpdesk.sla` `_name` must not `_inherit = 'helpdesk.sla'` unless Enterprise Helpdesk is in the registry (Chapter 2).

These 19 Enterprise addons have no 20 folder. Depend project / helpdesk / documents instead of the FSM bridges:

Odoo 19 Enterprise	Odoo 20
industry_fsm*	project (Chapter 14)
helpdesk_fsm / helpdesk_fsm_report / helpdesk_fsm_sale	helpdesk
documents_fsm	documents

**FIXED ASSETS — EE**

See Chapter 10. 20 adds account.depreciation.model . 19 templates were account.asset with state == 'model' .

**AI ADDONS — EE**

Enterprise 20 ships a set of ai\_\* modules. They are new optional apps, not a rename of a 19 module. Do not add them to a 19 graph unless the 20 database actually uses that stack.

**FLEET — CE + EE**

Odoo 19	Odoo 20
fleet.vehicle.log.services.date	date_from / date_to

---

# 18. Symptom index

---

[BACK TO CONTENTS](#)

Exception or result	Chapter
<code>KeyError: 'ir.rule'</code>	1
Field has no SQL representation	1
use <code>BinaryValue</code> instead of bytes	4
Incorrect padding / invalid <code>.xlsx</code>	4
No module named <code>xlwt</code> / <code>pytz</code>	4
<code>stock.scrap</code> missing	12
<code>hr.leave.type</code> / <code>hr.contract</code> missing	13
<code>hr.expense.sheet</code> missing	13
<code>industry_fsm</code> missing	14, 17
<code>crossovered.budget</code> missing	17
Invalid field <code>'company_type'</code> / <code>'groups_id'</code> / <code>'mobile'</code>	16
<code>mail.thread.cc</code> / MRO error	2
<code>useState</code> is not a function	9
Cannot read properties of undefined in Owl	9
<code>LiteralSQL</code> / <code>_select</code> concatenate	6
xpath parse / does not match	7
<code>&lt;act_window&gt;</code> / <code>&lt;report&gt;</code> RNG	8
<code>sale done</code> never matches	11
<code>portal /my</code> cards missing	14
<code>get_param</code> / <code>set_param</code> missing	2, 5
<code>check_access_rights</code> / <code>_name_search</code> missing	2, 3
request not bound	4
<code>type='json'</code> deprecation	5
ViewButton icon blank	7

Exception or result	Chapter
<code>_table_query</code> / overwrite <code>_table_sql</code>	6
<code>mail.tracking.value</code> missing	5, 15
<code>account.group</code> missing	10
<code>payment</code> <code>in_process</code> never matches	10
<code>show_produce_all</code> missing	12
<code>_get_profitability_items</code> missing	14
<code>helpdesk_fsm</code> / <code>documents_fsm</code> missing	17
<code>account.depreciation.model</code> missing on 19 code	10, 17
Font Awesome icon blank on website	7, 16
<code>request.website</code> / <code>env.website</code>	2, 16

# 19. Mapping tables

---

[BACK TO CONTENTS](#)

## MODELS

Odoo 19	Odoo 20	Edition
<code>hr.contract</code>	<code>hr.version</code>	CE + EE
<code>hr.leave.type</code>	<code>hr.work.entry.type</code>	CE + EE
<code>hr.expense.sheet</code>	<code>hr.expense</code>	CE + EE
<code>hr.payslip.input.type</code>	removed	EE
<code>stock.scrap</code>	<code>stock.move ( is_scrap=True )</code>	CE + EE
<code>stock.return.picking</code>	<code>stock.picking._create_return()</code>	CE + EE
<code>stock.quant.package</code>	<code>stock.package</code>	CE + EE
<code>note.note</code>	<code>project.task</code>	CE + EE
<code>mail.thread.cc</code>	<code>mail.thread</code>	CE + EE
<code>res.bank</code>	Char or <code>res.partner.bank</code>	CE + EE
<code>industry_fsm</code>	<code>project</code>	EE
<code>stock_picking_batch</code>	<code>stock</code>	CE + EE
<code>website_sale_wishlist</code>	<code>website_sale</code>	CE + EE
<code>website_sale_comparison</code>	<code>website_sale</code>	CE + EE
<code>crossovered.budget</code>	<code>budget.analytic</code>	EE
<code>crossovered.budget.lines</code>	<code>budget.line</code>	EE
<code>hr.version.wizard</code>	removed	CE + EE

Odoo 19	Odoo 20	Edition
<code>account.group</code>	<code>account.account ( parent_id )</code>	CE + EE
<code>account.asset</code> <code>state='model'</code>	<code>account.depreciation.model</code>	EE
<code>mail.tracking.value</code> (in <code>mail</code> )	<code>mail.tracking.value</code> (in <code>mail_tracking</code> )	CE + EE
<code>helpdesk_fsm</code> / <code>helpdesk_fsm_report</code> / <code>helpdesk_fsm_sale</code>	<code>helpdesk</code>	EE
<code>documents_fsm</code>	<code>documents</code>	EE

## FIELDS

Model	Odoo 19	Odoo 20	Edition
sale.order	state done	removed	CE + EE
stock.move.line / stock.move / purchase.order.line / mrp.bom / repair.order	product_uom_id / product_uom	uom_id	CE + EE
sale.order.line / account.move.line	product_uom_id	unchanged	CE + EE
res.partner.bank	acc_number , acc_holder_name	account_number , holder_name	CE + EE
res.partner	company_type , mobile	is_company , phone	CE + EE
res.users	groups_id	group_ids	CE + EE
ir.attachment	datas	raw	CE + EE
project.task	user_id	user_ids	CE + EE
product.template	type == 'product'	is_storable	CE + EE
hr.leave	holiday_status_id	work_entry_type_id	CE + EE
hr.leave	holiday_type	validation_type	CE + EE
account.analytic.account	group_id	plan_id	CE + EE
crm.lead	currency_id	company_currency	CE + EE
fleet.vehicle.log.services	date	date_from / date_to	CE + EE
payment.provider	state	is_live / is_published	CE + EE

Model	Odoo 19	Odoo 20	Edition
<code>resource.calendar.leaves</code>	<code>time_type</code>	<code>count_as</code>	CE + EE
<code>account.move</code> report	<code>type</code>	<code>move_type</code>	CE + EE
<code>account.account</code>	<code>group_id</code>	<code>parent_id</code> / <code>parent_ids</code>	CE + EE
<code>account.payment</code>	state <code>in_process</code>	<code>paid</code>	CE + EE
<code>account.payment</code>	state <code>paid</code>	<code>reconciled</code>	CE + EE
<code>mrp.production</code>	<code>show_produce_all</code>	<code>removed</code> ( <code>button_mark_done</code> )	CE + EE
Binary	bytes / base64	<code>BinaryValue</code> + <code>.filename</code>	CE + EE

## XMLIDS AND VIEWS

Odoo 19	Odoo 20
<code>sale.prod_config_main</code>	<code>sale.menu_sale_config</code>
<code>product.product_normal_action</code>	<code>product.product_variant_action</code>
<code>product.product_variant_easy_edit_view</code>	removed
<code>project.view_project_kanban</code> body	<code>project.view_project_card</code>
<code>project.view_task_kanban</code> card	<code>project.view_task_card</code>
<code>project.portal_my_home</code>	<code>portal.entry</code>
<code>industry_fsm.fsm_menu_root</code>	<code>project.menu_main_pm</code>
<code>hr_work_entry.work_entry_type_attendance</code>	<code>generic_work_entry_type_attendance</code>
pay-run kanban	<code>hr_payroll.hr_payrun_payslip_kanban_card</code>
Enterprise budget form	<code>account_budget.view_budget_analytic_form</code>
scrap form	<code>stock.view_scrap_move_form</code>
batch picking form	<code>stock.stock_picking_batch_form</code> in <code>stock</code>
<code>_table_query</code>	<code>_table_sql()</code> → <code>SQL</code>
<code>request.website</code> in models	<code>env.website</code>
wkhtml inside <code>web</code>	<code>base_report_wkhtmltox</code> / <code>base_report_paper_muncher</code>
JSON-2 <code>/json/2</code>	still present (preferred external API); XML-RPC present

# **Odoo Migration Cookbook: Version 19 to Version 20**

Community Edition and Enterprise Edition

First Edition, September 2026

Mr. Mustufa Rangwala and Ms. Sananaz Mansuri

Published by Probuse Consulting Service Pvt. Ltd.

419/A, SAKAR IX, Beside Old Reserve Bank of India,

Near City Gold Cinema, Ashram Road,

Ahmedabad 380009, Gujarat, India

Telephone: +91 96011 19434 · +91 78745 43092

Email: [contact@probuse.com](mailto:contact@probuse.com)

Website: <https://www.probuse.com>

Website: <https://probuseappdemo.com/>

Middle East and GCC

Contact: Ms. Sananaz Mansuri

Europe

Contact: Mr. Michel Rodriguez